

PRACTICAL COURSE WORKBOOK

AI & Algorithmic Trading

Build a clearer, auditable financial workflow

LEVEL	Advanced
GUIDED TIME	5 hours across 12 lessons
PROJECT	a fictional financial model with assumptions and checks
SKILLS	Algo trading, ML, Backtesting, QuantLib, Signals, Critical evaluation
EDITORIAL STATUS	Structured draft v0.9

WHAT YOU WILL BE ABLE TO DO

- Use Algo trading appropriately in a realistic, bounded task.
- Use ML appropriately in a realistic, bounded task.
- Use Backtesting appropriately in a realistic, bounded task.
- Use QuantLib appropriately in a realistic, bounded task.

WHO THIS IS FOR

Learners studying financial technology and analytical methods.

BEFORE YOU BEGIN

- Basic percentages and spreadsheet skills

This open workbook is a structured draft undergoing course-specific editorial review. It does not provide certification.

COURSE MAP

From foundation to finished work

AI & Algorithmic Trading is a structured, practical course covering Algo trading, ML, Backtesting, QuantLib, Signals. It emphasizes explainable methods, checked work and a final outcome that can be reviewed.

MODULE	LESSONS AND FOCUS
01 Algo trading: Financial foundations	1. Value with Algo trading 2. Risk with ML 3. Markets with Backtesting
02 ML: Tools and data	1. Records with QuantLib 2. Analysis with Signals 3. Controls with Algo trading
03 Backtesting: Applied decisions	1. Scenarios with ML 2. Evaluation with Backtesting 3. Reporting with QuantLib
04 QuantLib: Risk and governance	1. Security with Signals 2. Regulation with Algo trading 3. Review with ML

HOW TO STUDY EACH LESSON

- 1 Read the objective and identify the decision it supports.
- 2 Reproduce the worked example with the stated input.
- 3 Test one normal case and one boundary or failure case.
- 4 Complete the knowledge check without guessing.
- 5 Save a short evidence note: result, limitation and next action.

CAPSTONE PROJECT

Produce a reviewable AI & Algorithmic Trading project using Algo trading, ML, Backtesting.

DELIVERABLE	A working AI & Algorithmic Trading artefact plus an evidence-based self-review.
TOOLS	A suitable Algo trading environment, A plain-text decision log, Test data or realistic sample material

Production plan

- 1 Define the intended user, outcome and constraints.
- 2 Create the smallest complete result using Algo trading.
- 3 Test one normal case, one boundary case and one failure response.
- 4 Revise the work from the evidence and preserve before-and-after results.
- 5 Prepare a concise handover containing method, limitations and next step.

Definition of done

- The output matches the stated outcome.
- Inputs and decisions are reproducible.
- Boundary and failure evidence is included.
- Limitations and responsibility considerations are explicit.

RESPONSIBLE PRACTICE

This course is general education, not financial, investment, tax or legal advice.

Evidence log

INPUT	Original brief, data, requirements or test material.
DECISIONS	Chosen method and one reasonable alternative.
TESTS	Normal, boundary and failure results.
LIMITATION	What the result does not establish.
REVISION	One evidence-led improvement.

REVIEW AND SOURCES

Make the work credible and repeatable

Final self-review

- Can another learner repeat the method?
- Did I test a difficult case?
- Did I avoid unsupported claims?
- Is the next action proportionate to the remaining risk?

Authoritative references

National Payments System

Central Bank of Kenya - reviewed 2026-08-21
<https://www.centralbank.go.ke/national-payments-system/>

Financial education

OECD - reviewed 2026-08-21
<https://www.oecd.org/en/topics/financial-education.html>

SOURCE NOTE

Product versions, laws and professional standards can change. Recheck the linked primary source before applying version-sensitive information.

NEXT IMPROVEMENT

Choose one weakness found during review and improve it before extending the Algo trading scope.